

Touch

In neuroscience, touch refers to the special sense by which contact with the body is perceived consciously.

Active & passive touch stimulate the same receptors.

The hand has 4 types of mechanoreceptors

Touch: combined result of information of these 4.

functional unit = afferent fiber, the fiber's distant branches, and the receptor organ that surrounds the axon terminal.

Rapid & Slow, 1 & 2 →

surface → deeper



← Action potentials
← Stimulus

↑
encapsulated → detects motion (detect up to 30-500 Hz)

RA1 → Meissner - globules, filled with fluid, originates from myelin sheath.

An RA1 axon → 10-20 Meissner corpuscles

2-5 " " " 1 " "

✓ frictional force.

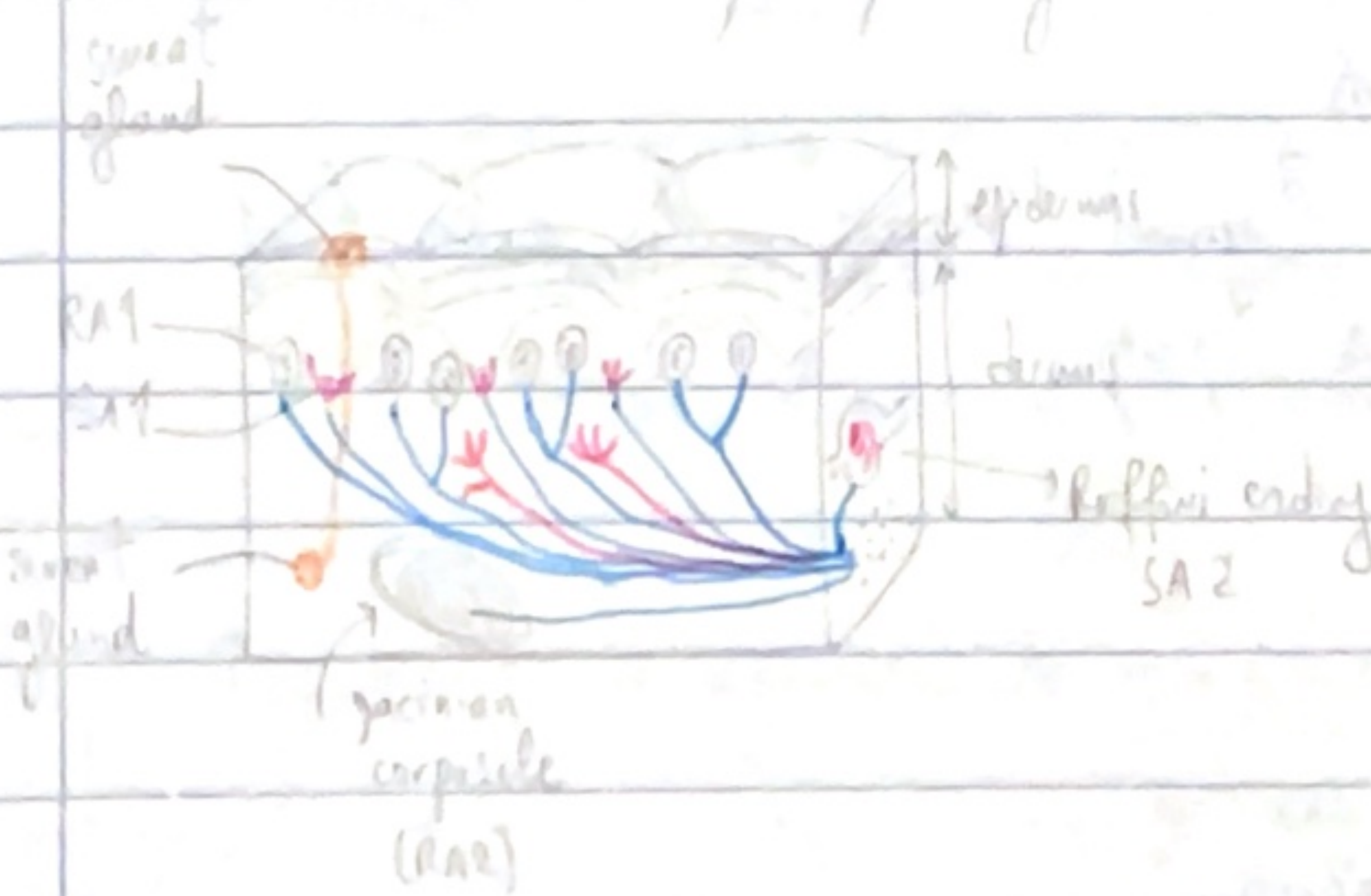
SA1 → well distributed, particularly in the fingertips
epithelial cells (MERKEL CELLS)

✓ compressive strain

densely clustered in the center of each papillary ridge in

glabrous skin

in palm / on glabrous skin



≠ The fingerprints give the glabrous skin a convoluted, rough structure that increases friction allowing us to grasp objects without slippage.

dermatoglyphs

Type 2 = deeper, innervate the skin sparsely - receptors are larger and less numerous.
↓
sense displacement

RA2 $\rightarrow$ Jansen corpuscle $\rightarrow$ (1) Axon. - Subcutaneous tissue.
(unmyelinated)

Sensitive high frequency vibration (2400 pps - 300 elderly)

SA2 $\rightarrow$ Ruffini endings - finger and wrist joints and along skin folds (rare in fingertips)

(+) by stimuli that stretch the receptor along its long axis

$\Rightarrow$ Receptive field. - smallest at fingertips and lips.

threshold for spatial acuity $\rightarrow$ 1mm for young people } discrimination
2mm old.

Slowly adaptive $\rightarrow$ Deformation and Pressure. - edges, joints, curvatures.



soft if skin "changes" object

hard if object " " skin

Shallow
sloped
basin



local region also governed by
basal curvature $\rightarrow$ TENSILE STRAIN

(If I press with another finger on the
sloped basin, the force is $\downarrow$)

"less is more"

finger 1 $\rightarrow$ symmetric

" 2 $\rightarrow$ a "

S2 $\rightarrow$ Palmer folds / finger joints.

$\rightarrow$ shape of large objects, Palpate objects with entire hand
info about hand while empty

3D (stereognosis)

proprioceptive

Rapid adaptive $\rightarrow$ Motion and vibration

RA1 $\rightarrow$ irregularities and bumps as small as 10 μ m

RA2 $\rightarrow$ most sensitive mechanoreceptor
detects vibration

(Micro neurography)

All 4 classes respond to grasp.

SA1 - amount of grasp force by each finger

RA1 - how quickly the grasp is applied } if grasp is established they "turn off"

2 - detects lifting

SA2 - flexion / extension fingers.

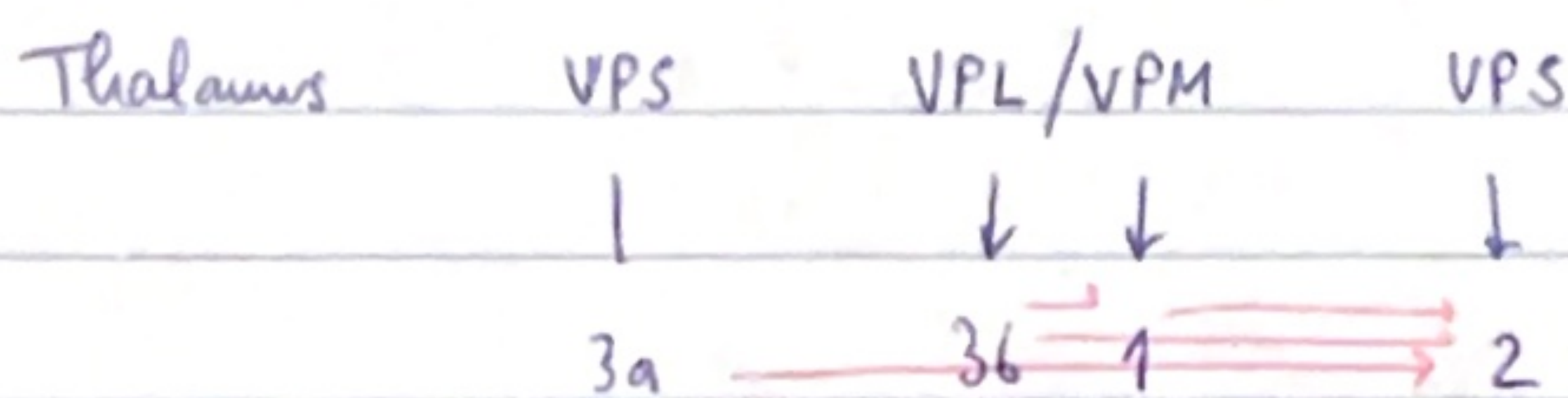
Somatosensory areas:

SI 3a, 3b, 1, 2

SII 43

~~SIII~~ S and 7, 39, 40

Posterior parietal cortex 5-7 somatosensory
39-40 " and visual



3a → Muscle Spindles — deep tissue

3b → SA1, RA1 receptors

1 → RA1, RA2 receptors

2 → Complex touch, proprioception

5 → Active touch

⇒ Stimulus is highest in the middle of receptive field

• Some neurons only excitatory (6%), others are regulated by inhibition (42%)
and others are excited by stimulation in an excitatory zone and
then inhibition in the surrounding inhibitory zone.

The cortex is organized in columns → anatomical structure
governs one submodality